FinNews Intelligence

A Streamlit Text Mining & Document Intelligence System for Financial Newswire Analysis

AIMLZG537 - Information Retrieval — Text Mining Assignment

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Headline results	
Corpus	372 Reuters-21578 stories, 6 topics (357 after de-duplication)
Naive Bayes (5 seeds)	accuracy 0.861 ± 0.035, macro-F1 0.862 ± 0.035
K-Means, K = 6 (5 seeds)	Purity 0.665 ± 0.073, Rand Index 0.833 ± 0.030
Recommendation	content-based Top-K; precision@5 = 0.721
Duplicates	26 pairs found (2 exact, 13 near, 11 template); all 12 injected copies detected

1. Selected business context

An **investment-research desk** at an asset manager receives a continuous stream of financial newswire stories. Analysts need every story **tagged** with a market topic so it reaches the right desk (FX, rates, energy, M&A, trade policy, shipping); they want to **discover** themes that cut across the editorial taxonomy; they lose time on **re-issued and corrected copies** of the same story; and when reading a story they want the **related coverage** at hand. The system built here addresses each need: manual Multinomial Naive Bayes for tagging, manual K-Means for theme discovery, shingle/cosine duplicate detection, and a content-based Top-K recommender, all driven from a Streamlit front end.

2. Dataset description

The corpus is drawn from **Reuters-21578** (ApteMod distribution, NLTK), the standard benchmark of real 1987 Reuters financial newswire. `data/prepare_dataset.py` selects stories that carry exactly one topic label and at least 80 words, samples ~60 per topic with a fixed seed, separates the headline from the body, and writes one CSV in which text and metadata are separate columns.

Topic	Desk	Documents	Mean words	Median words
acq	Mergers & Acquisitions	61	159	114
crude	Crude Oil	61	195	141
interest	Interest Rates	60	184	124
money-fx	Foreign Exchange	66	268	165
ship	Shipping	64	145	110
trade	International Trade	60	218	180
Total		**372**	195	144

- **Size:** 372 documents, 72,681 words, 0.47 MB of body text (requirement: at least 50-75 documents).
- **Fields:** *doc_id* (unique, e.g. RTR-06301), *title*, *text*, *category* (label), *category_name*, *source*, *split* (original ApteMod train/test), *word_count*, *injected_dup_of*. The dashboard and ingestion page display metadata and text separately.
- **Natural noise:** all-capital headlines, HTML entities and ticker markup (`<GY>`), newswire jargon (*mln, dlrs, pct, bpd, stg*), tables of figures, inconsistent spacing.
- **Duplicates:** the corpus contains genuine exact duplicates and re-issued stories, plus templated weekly reports. To make duplicate handling measurable, 12 synthetic near-duplicates (DUP-001..012) were added that mimic real syndication patterns: an 'UPDATE 1-' re-issue with the last sentence dropped, an upper-cased copy wrapped in HTML, and a 'CORRECTED' copy with repeated sentences and punctuation noise. They are flagged in *injected_dup_of*.
- **Supports every task:** labels for Naive Bayes and for Purity/Rand Index, and text for content-based recommendation. Collaborative filtering is not used because the corpus has no analyst-story interaction log (Section 9.3).

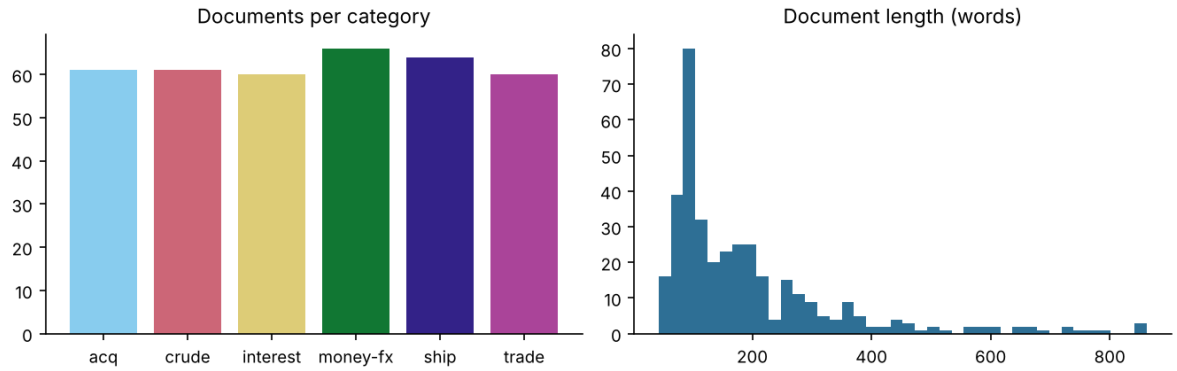


Figure 1 - Category distribution and document length.

3. Implementation approach

The code is split into a small library (`textmining/`) and the Streamlit front end (`app.py`). Every scored algorithm is written by hand; numpy is used only for array arithmetic and NLTK only for its stop-word list, Porter stemmer and WordNet lemmatizer. scikit-learn is not imported.

Module	Implemented manually
preprocess.py	markup stripping, regex tokenizer, lowercasing, number removal, generic + domain stop-words, stemming / lemmatization, n-gram generation, minimum token length
features.py	inverted index (term -> postings [(doc, tf)]), df / idf, TF-IDF weights with cosine length normalisation, cosine similarity, term-at-a-time query scoring through the postings
classify.py	Multinomial Naive Bayes with Laplace smoothing in log space, stratified (duplicate-group aware) train/test split, confusion matrix, precision / recall / F1, macro and weighted averages
cluster.py	K-Means with k-means++ or random seeding, logged assignment / update steps, two stopping rules, multiple restarts; Purity; Rand Index over all document pairs; PCA via SVD (display only)
dedup.py	MD5 for exact duplicates, word 5-shingle Jaccard and TF-IDF cosine for near/template duplicates, union-find grouping
recommend.py	content-based Top-K by cosine, duplicate exclusion, term-level explanations, precision@K
experiments.py	controlled comparisons over preprocessing settings, de-duplication modes and K, averaged over random seeds

Streamlit workflow. Six pages in the sidebar follow the analyst's workflow: *Dashboard* (corpus overview and headline metrics) → *Data ingestion* (bundled corpus or upload of CSV / JSON / ZIP of .txt with column mapping, ID-uniqueness check and corpus statistics) → *Preprocessing & features* (configuration form, before/after vocabulary, inverted-index explorer, TF-IDF vectors, similarity heat-map) → *Text mining hub* (Naive Bayes, K-Means with every step, evaluation and comparison, duplicates) → *Recommendations* → *Performance analytics* (re-runs the pipeline under alternative settings). Results are cached per configuration, so changing a setting re-runs only what is needed. Screenshots are in the Appendix.

4. Preprocessing and feature choices

Chosen configuration (E): strip markup and decode entities → regex tokenisation → lowercase → drop numbers → remove English stop-words plus newswire boiler-plate (*said, mln, dlrs, pct, reuter, inc, corp*) → WordNet lemmatization (noun then verb) → tokens of length ≥ 2 → unigrams. The inverted index keeps terms with df ≥ 2 and df ≤ 90% of documents. Section 9.1 justifies this choice experimentally.

Statistic	Before (raw tokens)	After preprocessing	Change
Vocabulary size	8,793	5,276	-40.0%
Total tokens	71,061	37,798	-46.8%
Hapax legomena (freq = 1)	4,291	2,203	-48.7%
Mean / median tokens per doc	199 / 144	106 / 74	
Indexed terms (df >= 2, df <= 90%)		2,609	2,667 pruned

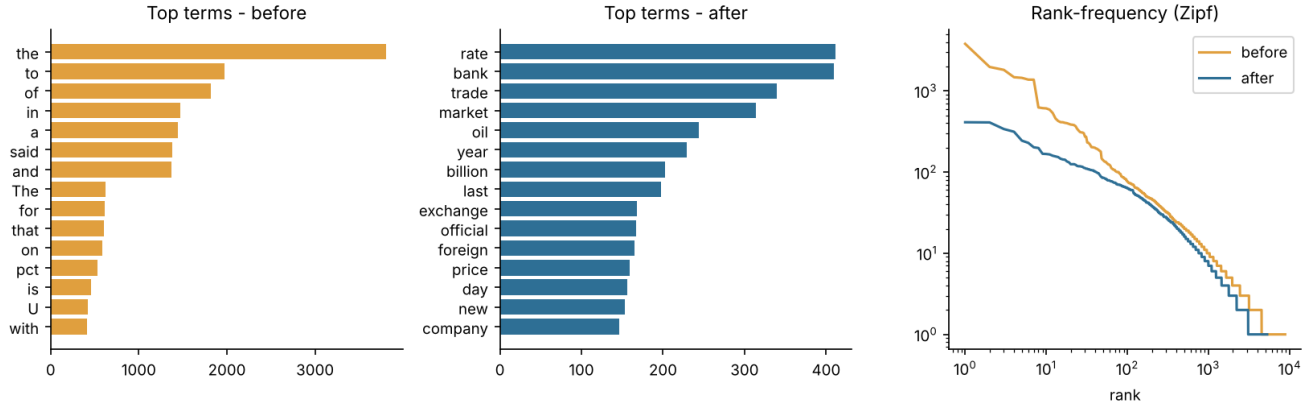


Figure 2 - Most frequent terms before and after preprocessing, and the rank-frequency (Zipf) curves. Before, the head is dominated by function words and 'said', 'mln', 'dlrs'; after, by topical terms (oil, rate, trade, bank, share).

TF-IDF (SMART ltc), computed from the inverted index without libraries. For term t in document d:

$$w(t,d) = (1 + \log_{10} tf_{t,d}) \times \log_{10}(N / df_t), \quad w'(t,d) = w(t,d) / \mathrm{sqrt}(\Sigma_{t'} w(t',d)^2)$$

The postings lists provide tf and df; idf is computed once per term; each document vector is then divided by its Euclidean length so every story has unit length regardless of how long it is. Document similarity is the cosine, cos(q,d) = q·d / (|q||d|), which reduces to a dot product for unit vectors. Free-text queries are vectorised with the corpus idf and scored term-at-a-time by walking only the postings of the query terms.

5. Classification results - Multinomial Naive Bayes

The model estimates log P(c) and log P(t|c) = log[(count(t,c) + α) / ($\Sigma_{t'}$ count(t',c) + α|V|)] with α = 1 and predicts argmax$_c$ [log P(c) + Σ_t tf(t,d) log P(t|c)]. The split is stratified 70/30 and keeps every duplicate group on one side, giving 249 training and 108 test documents. Results below are for the default seed (42); averaged over 5 seeds the accuracy is **0.861 ± 0.035** and macro-F1 **0.862 ± 0.035**, so seed 42 sits at the lower end.

Class	Precision	Recall	F1	Support
acq	0.895	0.944	0.919	18
crude	0.882	0.833	0.857	18
interest	0.765	0.722	0.743	18
money-fx	0.636	0.778	0.700	18
ship	0.929	0.722	0.813	18
trade	0.789	0.833	0.811	18
Macro avg	0.816	0.806	0.807	108
Weighted avg	0.816	0.806	0.807	108
Accuracy			**0.806**	108

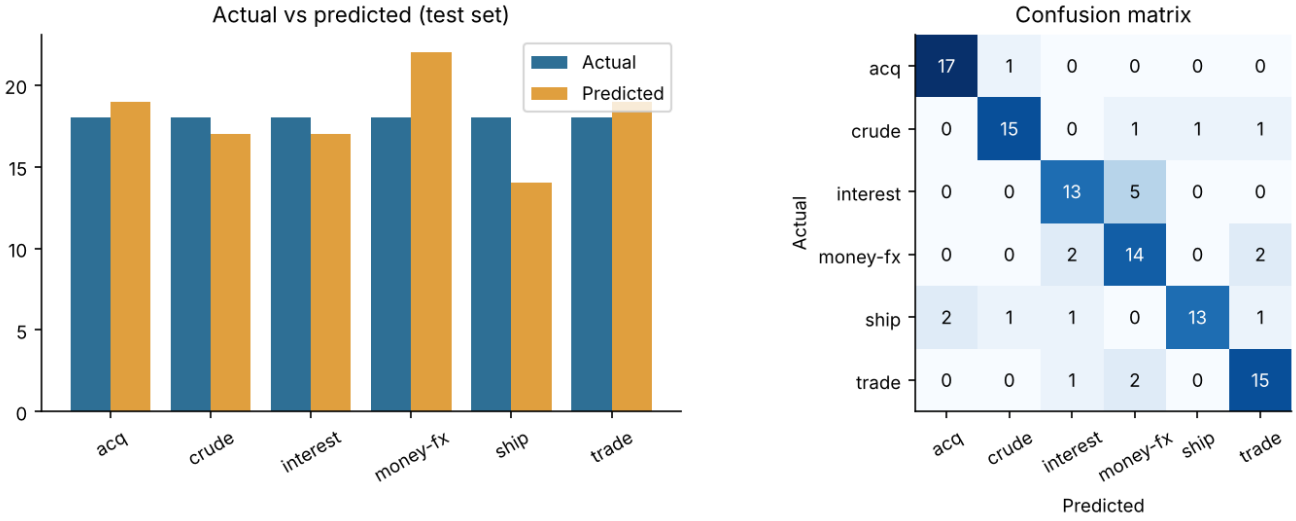


Figure 3 - Actual vs predicted class distribution on the test set, and the confusion matrix (rows = actual).

Reading the errors. *acq* is the easiest class (F1 0.919): takeover stories share a distinctive vocabulary (stake, acquisition, shareholder). The hardest pair is *interest* vs *money-fx*: 5 interest stories were predicted as money-fx and 2 money-fx stories as interest, because central-bank stories discuss both rates and currencies. *ship* has high precision but lower recall; some shipping stories about Gulf tankers and port strikes are pulled towards *crude*.

Sample predictions (test set; all predictions are listed in the app and in results.json):

doc_id	Title	Actual	Predicted	
RTR-10625	ISLAMIC BANKS ESTABLISH 50 MLN DLR TRADE PORTFOLIO	trade	money-fx	wrong
RTR-08098	U.S. CREDIT MARKET OUTLOOK - MINI-REFUNDING	money-fx	interest	wrong
RTR-03449	SUMITA SAYS LITTLE ROOM FOR BANK TO EASE POLICY	interest	money-fx	wrong

doc_id	Title	Actual	Predicted	
RTR-03046	HUTCHISON UNIT BUYS CONTAINER CRANES FROM JAPAN	ship	acq	wrong
RTR-07310	GERMAN BANKS SEE LOW INTEREST RATES CONTINUING	interest	money-fx	wrong
RTR-00281	CHINA SIGNS WORLD BANK LOAN FOR TIANJIN PORT	ship	crude	wrong
RTR-16745	MIYAZAWA SEES EVENTUAL LOWER U.S. TRADE DEFICIT	trade	money-fx	wrong
RTR-07500	U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS	ship	ship	correct
RTR-05796	STUDY PREDICTS U.S. DEPENDENCE ON FOREIGN OIL	crude	crude	correct
RTR-01499	BALDRIGE CONCERNED ABOUT KOREAN/TAIWAN DEFICITS	trade	trade	correct
RTR-11738	LILLY CUTS LIPOSOME CO STAKE	acq	acq	correct
RTR-04125	CHINA CLOSES SECOND ROUND OF OFFSHORE OIL BIDS	crude	crude	correct
RTR-06483	BANK OF ENGLAND LENDS ONE BILLION STG AT 10 PCT	interest	interest	correct
RTR-16096	YIELD RISES ON 30-DAY SAMA DEPOSITS	interest	interest	correct

Most discriminative terms per class (highest log P(t|c)/P(t|not c)): *acq*: acquisition, stake, partnership, keycorp, taft; *crude*: barrel, bpd, opec, crude, energy; *interest*: thai, three-month, eurosterling, maturity, borrower; *money-fx*: stoltenberg, poehl, louvre, em, employment; *ship*: ship, port, seaman, vessel, redundancy; *trade*: gatt, yeutter, tariff, poland, sanction.

6. Clustering results - K-Means

K-Means runs on the unit-length TF-IDF vectors with squared Euclidean distance (for unit vectors $||x - c||^2 = 2 - 2\cos$, so it groups by cosine similarity). K is set to the number of topics (6). The four steps of the algorithm are logged and shown in the app:

- **Initialisation (k-means++):** the first seed is a random document; each next seed is sampled with probability proportional to its squared distance from the nearest seed already chosen. Seeds of the best run: RTR-04425, RTR-20262, RTR-12040, RTR-11349, RTR-15384, RTR-20653.
- **Assignment:** every document goes to the centroid with the smallest squared distance.
- **Update:** each centroid becomes the mean of its member vectors (an empty cluster would be re-seeded with the farthest document).
- **Stopping:** stop when no document changes cluster, or the largest centroid shift is below 10^{-6}, or after 100 iterations. The best run stopped after 12 iterations because no document changed cluster (assignments converged). Five restarts were made and the lowest-SSE run kept (SSEs: 330.6, 327.6, 325.4, 326.4, 329.7).

Iteration	Docs reassigned	Max centroid shift	SSE	Cluster sizes
1	6	0.94172	331.65	[65, 88, 20, 58, 91, 35]
2	33	0.15543	329.70	[75, 87, 15, 53, 90, 37]
3	20	0.30468	328.14	[78, 90, 9, 52, 90, 38]
4	8	0.22278	327.34	[79, 91, 7, 50, 91, 39]
5	8	0.17046	326.88	[82, 87, 6, 49, 94, 39]
6	5	0.02609	326.73	[83, 88, 6, 49, 92, 39]
7	9	0.03902	326.45	[83, 83, 6, 49, 97, 39]
8	10	0.04379	326.13	[85, 77, 6, 50, 100, 39]
9	11	0.04529	325.77	[87, 75, 6, 50, 99, 40]
10	7	0.03877	325.56	[88, 70, 6, 50, 103, 40]
11	7	0.03293	325.37	[86, 67, 6, 51, 107, 40]
12	0	0.00000	325.37	[86, 67, 6, 51, 107, 40]

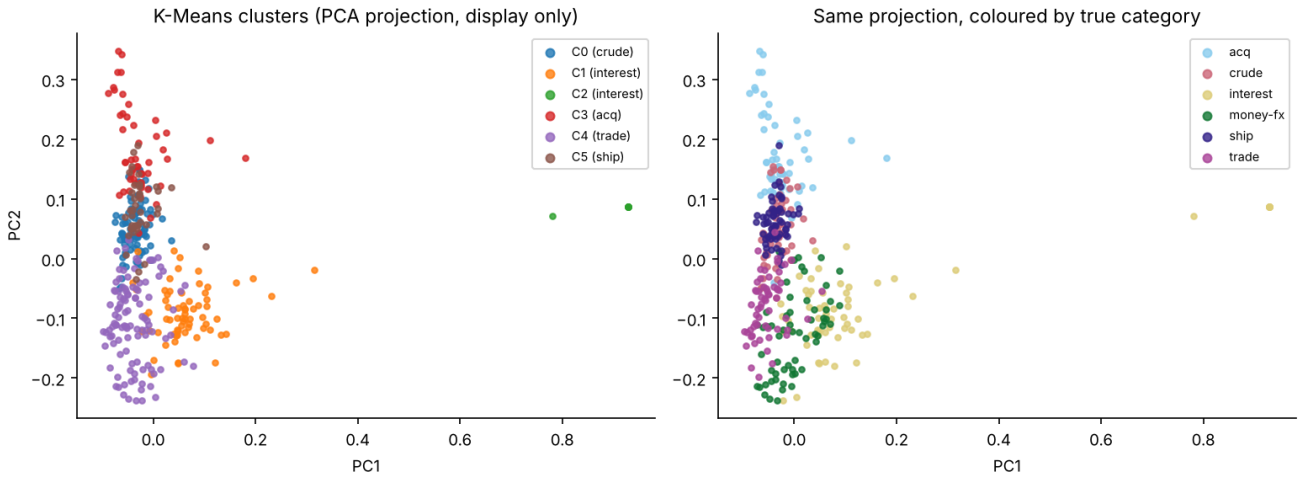


Figure 4 - Documents projected to 2-D with PCA (visualisation only, PCA computed by numpy SVD; the first two components explain 3.2% of variance). Left: K-Means clusters; right: true topics.



Figure 5 - Cluster sizes, cluster x topic contingency table, and convergence of SSE / reassignments.

Cluster	Size	Majority topic (share)	Top centroid terms
C0	86	crude (49%)	oil, crude, barrel, ship, port, bpd, strike, opec
C1	67	interest (64%)	bank, rate, stg, lend, money, prime, point, bill
C2	6	interest (100%)	maturity, old, day, short-term, adjust, discount, loan, home
C3	51	acq (96%)	share, stake, stock, common, company, acquire, group, acquisition
C4	107	trade (50%)	trade, japan, currency, economic, country, foreign, exchange, nation
C5	40	ship (40%)	sea, water, canadian, continental, sink, drill, vessel, oil

Evaluation. Purity = (1/N) $\Sigma_k \max_j |\omega_k \cap c_j|$ = **0.588**. Rand Index = (TP + TN)/(TP + FP + FN + TN) over all 63,546 pairs (TP = 6,386, FP = 7,221, FN = 4,057, TN = 45,882) = **0.823**. Both are computed by hand-written functions. Over 5 seeds: Purity 0.665 ± 0.073, RI 0.833 ± 0.030.

What the clusters found. C3 is an almost pure M&A cluster. C0 mixes *crude* (42) and *ship* (41): in 1987 much shipping news was about tankers attacked in the Gulf and oil-port strikes, so the unsupervised view sees one 'oil and tankers' theme where the taxonomy has two. C1 joins *interest* and *money-fx* (central banks), and C4 joins *trade* with *money-fx* (the US-Japan trade dispute and the yen). C2 is a 6-document cluster of short-term discount-note rate tables (five FHLBB, one Sallie Mae): these templated reports are near-identical after number removal and capture a centroid on their own (see Section 9.2). C5 collects the remaining shipping and offshore stories.

Cluster membership (first 14 documents; the full table is in the app and results.json):

doc_id	Title	True label	Cluster	Cluster majority
RTR-07500	U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS	ship	C5	ship
RTR-11161	PARTNERS CALL GENCORP RESPONSE UNPRODUCTIVE	acq	C3	acq

doc_id	Title	True label	Cluster	Cluster majority
RTR-03622	CONTINENTAL HEALTH PURCHASES MARKETECH	acq	C5	ship
RTR-03256	LONDON GRAIN FREIGHTS	ship	C5	ship
RTR-20311	BASTIAN TECHNOLOGIES MULLING COSMO BID	acq	C3	acq
RTR-21343	FED SEEN ADDING RESERVES VIA SYSTEM REPOS	money-fx	C1	interest
RTR-05796	STUDY PREDICTS U.S. DEPENDENCE ON FOREIGN OIL	crude	C5	ship
RTR-21131	AMPLE SUPPLIES LIMIT U.S. STRIKE'S OIL PRICE IMPACT	crude	C0	crude
RTR-02542	TALKS FAIL TO END BRAZILIAN SEAMEN'S STRIKE	ship	C0	crude
RTR-01499	BALDRIGE CONCERNED ABOUT KOREAN/TAIWAN DEFICITS	trade	C4	trade
RTR-17980	LAWSON SAYS LOUVRE CURRENCY ACCORD SATISFACTORY	money-fx	C4	trade
RTR-11738	LILLY CUTS LIPOSOME CO STAKE	acq	C3	acq
RTR-10625	ISLAMIC BANKS ESTABLISH 50 MLN DLR TRADE PORTFOLIO	trade	C4	trade
RTR-20039	LAWSON SAYS BP SHARE OFFER GOING AHEAD	acq	C1	interest

7. Comparison of classification and clustering

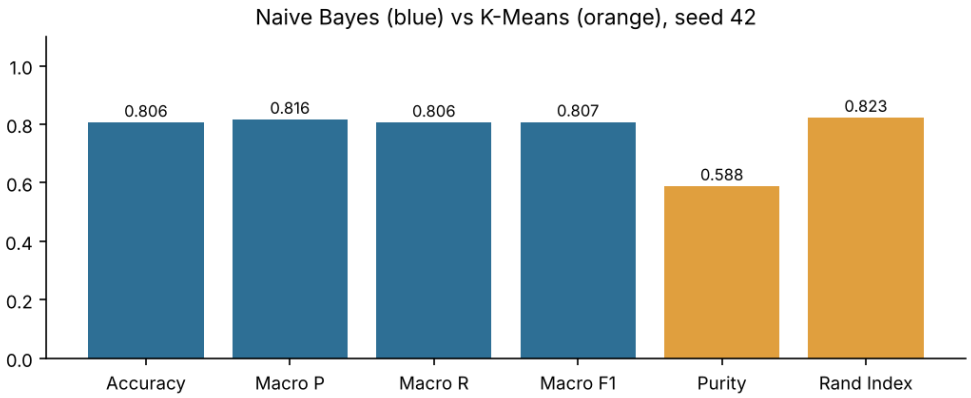


Figure 6 - Naive Bayes metrics and K-Means Purity / Rand Index (seed 42).

	Naive Bayes (supervised)	K-Means (unsupervised)
Uses labels?	Yes - 249 labelled training stories	No - labels only used afterwards for evaluation
Metrics	Accuracy, precision, recall, F1, confusion matrix	Purity, Rand Index, contingency table
Seed 42	Accuracy 0.806, macro-F1 0.807	Purity 0.588, RI 0.823
Mean ± std, 5 seeds	Accuracy 0.861 ± 0.035, macro-F1 0.862 ± 0.035	Purity 0.665 ± 0.073, RI 0.833 ± 0.030
Reference point	Majority-class guess = 17%	One cluster: purity 17%; random 6-way split: RI about 0.72
Main confusion	interest vs money-fx	crude + ship merged; interest + money-fx merged
Stability	std 0.035 (split only)	std 0.073 (seeding + split of local minima)

The supervised classifier is clearly better at reproducing the editorial taxonomy, as expected: it is told what the classes mean. K-Means has to infer structure from word co-occurrence alone; it recovers the M&A topic almost perfectly, but it groups by *theme* rather than by desk, merging topics whose stories share vocabulary. Purity and accuracy are not directly comparable (purity lets several clusters share a majority label and ignores small classes), and the Rand Index looks high partly because most pairs are 'different class, different cluster' (TN = 45,882 of 63,546); against a random baseline of about 0.72 the gain is real but moderate. The two methods are complementary: Naive Bayes for routing stories into a fixed taxonomy, K-Means for spotting themes that cut across it.

8. Recommendation results

Approach: content-based Top-K. A query story (or a free-text description) is represented by its unit-length TF-IDF vector; every other story is scored by cosine similarity and the K highest are returned with their score, topic, source and the terms that contribute most to the dot product. The query's own duplicates (same duplicate group, or cosine ≥ 0.90) are excluded, because they would otherwise occupy the top slots. Free-text queries are scored through the inverted index, touching only documents that share a query term.

Query RTR-06301 - *API SAYS DISTILLATE, GAS STOCKS OFF IN WEEK* (topic: crude)

#	doc_id	Title	Topic	Cosine	Top shared terms
1	RTR-16005	CRUDE OIL NETBACKS UP SHARPLY IN EUROPE, U.S.	crude	0.202	gasoline, refinery, distillate, fell
2	RTR-08209	STUDY SAYS OIL PRICE FALL SPURS BANKRUPTCIES	crude	0.184	api, institute, american, petroleum
3	RTR-00273	SAUDI FEBRUARY CRUDE OUTPUT PUT AT 3.5 MLN BPD	crude	0.174	refinery, storage, bpd, fell
4	RTR-04425	MONTEDISON, OTHER REFINERIES TO SHUT FOR WORK	crude	0.168	refinery, capacity, bpd, crude
5	RTR-13266	IEA FORECASTS SLOWER GROWTH IN OECD OIL DEMAND	crude	0.151	fuel, bpd, fell, stock

Query RTR-08710 - *SUMITA SAYS BANK WILL INTERVENE IF NECESSARY* (topic: money-fx)

#	doc_id	Title	Topic	Cosine	Top shared terms
1	RTR-03433	MIYAZAWA SAYS PARIS ACCORD HELPING STABILISE R	money-fx	0.212	paris, cooperate, france, britain
2	RTR-03449	SUMITA SAYS LITTLE ROOM FOR BANK TO EASE POLIC	interest	0.206	sumita, satoshi, yen, governor
3	RTR-11430	TREASURY'S BAKER SAYS U.S. BACKS STABILITY	money-fx	0.204	consistent, broadly, paris, cooperate
4	RTR-09797	TREASURY'S BAKER SAYS HE STANDS BY PARIS PACT	money-fx	0.203	consistent, broadly, paris, yen
5	RTR-16770	JAPAN'S LDP TO CALL FOR FLEXIBLE MONETARY POLI	money-fx	0.198	sumita, satoshi, necessary, adequate

Query RTR-11349 - *KEYCORP AGREES TO ACQUIRE UTAH BANK* (topic: acq)

#	doc_id	Title	Topic	Cosine	Top shared terms
1	RTR-17770	U.S. BANCORP HAS ACQUISITION APPROVAL	acq	0.278	bancorp, salt, conversion, definitive
2	RTR-07635	C.O.M.B. TO ACQUIRE REST OF CABLE VALUE	acq	0.177	subject, shareholder, transaction, proposal
3	RTR-16277	BOSTON FIVE , NEWORLD TO MERG	acq	0.172	bancorp, merger, become, asset
4	RTR-06856	LDBRINKMAN CHIEF CUTS CENERGY STA	acq	0.164	family, share, outstanding, common
5	RTR-02219	HOLDER PROPOSES UNITEL SALE OR LIQUIDATI	acq	0.162	shareholder, merger, common, outstanding

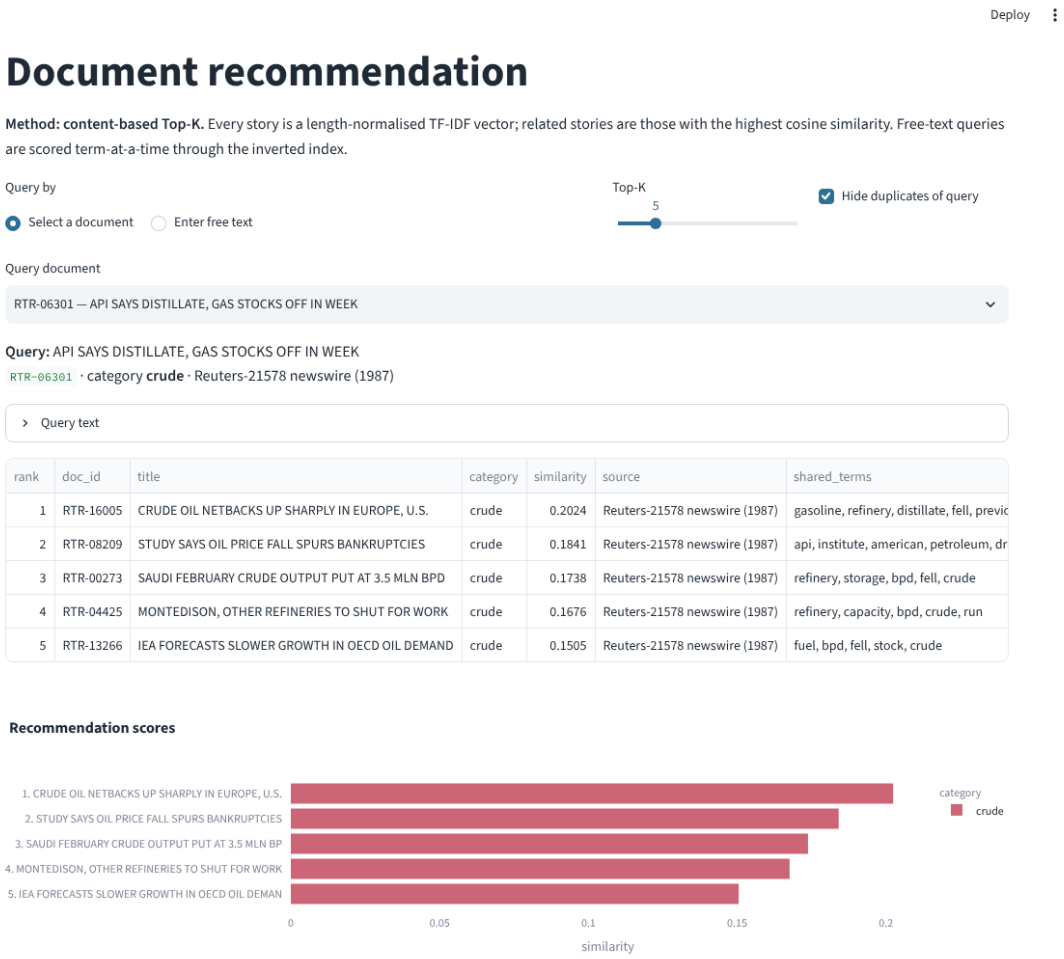


Figure 7 - Recommendation interface: query selection, Top-K slider, duplicate filter, result table with scores and metadata, and score chart

Findings.

- Across the whole corpus, 72.1% of Top-5 recommendations share the query's topic (precision@5, used as an offline proxy since no relevance judgements exist). For the three queries above, 14 of 15 recommendations are on-topic, and the one 'miss' (RTR-03449, labelled *interest*, for a Bank of Japan intervention story) is in fact highly relevant - same speaker, same policy - which shows the label-based proxy understates usefulness.
- Without the duplicate filter, the top hit for RTR-06301 would be RTR-01343 (cosine 0.970): the previous week's API bulletin, which uses identical wording with different figures. Every one of the 30 documents that has a duplicate would otherwise get that duplicate as its top recommendation.
- Scores for genuinely related but distinct stories are low in absolute terms (0.15-0.30) because short wire stories share few terms; ranking, not the absolute value, is what matters.

Limitations on this dataset.

- Purely lexical: synonyms and paraphrases ("intervene" vs "buy dollars") are missed; there is no semantic model.
- Templated reports look identical once numbers are removed; the recommender cannot tell this week's figures from last week's.
- No recency, source authority or personalisation; every analyst receives the same list.
- Short stories have sparse vectors, so a few shared names (Sumita, Paris) can dominate the score.

9. Inference and discussion

All experiments re-run the complete pipeline and report the mean (and standard deviation) over 5 random seeds, which change both the train/test split and K-Means seeding. This matters: on a 108-document test set a single split

can move accuracy by ±4 points.

9.1 How did preprocessing affect vocabulary, classification, clustering and recommendation?

	Steps	Vocab	Index terms	NB accuracy	NB macro-F1	Purity	Rand Index	Rec P@5
A	Raw tokens (no cleaning)	8,969	3,885	0.813 ± 0.017	0.814 ± 0.017	0.557 ± 0.055	0.788 ± 0.028	0.711
B	Lowercase + markup/numbers removed	6,973	3,349	0.809 ± 0.045	0.810 ± 0.047	0.648 ± 0.047	0.821 ± 0.020	0.724
C	B + stop-words (generic + domain)	6,822	3,217	0.852 ± 0.041	0.853 ± 0.040	0.617 ± 0.018	0.814 ± 0.008	0.725
D	C + Porter stemming	4,919	2,477	0.863 ± 0.032	0.864 ± 0.032	0.593 ± 0.037	0.806 ± 0.018	0.716
E	C + WordNet lemmatization	5,276	2,609	0.861 ± 0.035	0.862 ± 0.035	0.665 ± 0.073	0.833 ± 0.030	0.721
F	E + bigrams	35,285	5,270	0.848 ± 0.022	0.851 ± 0.023	0.558 ± 0.056	0.786 ± 0.032	0.723

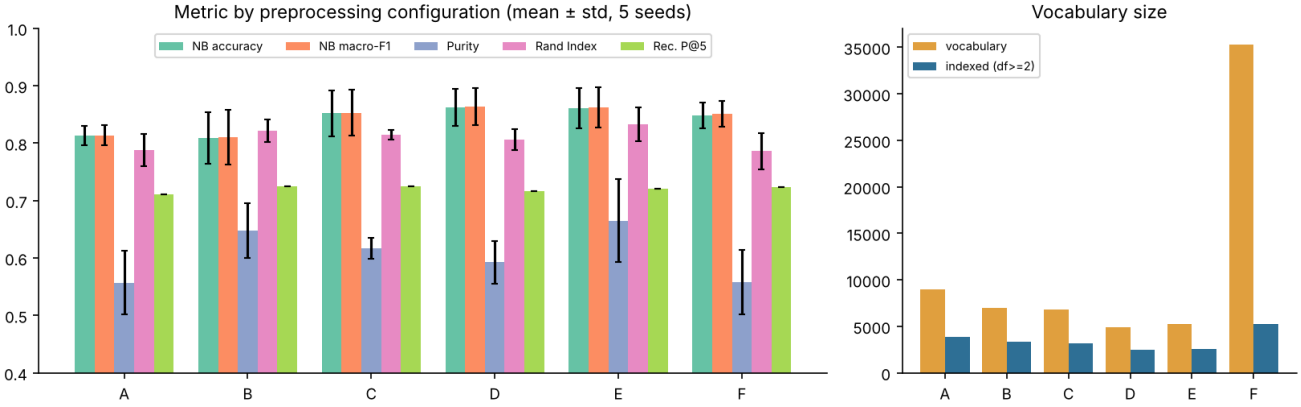


Figure 8 - Effect of each preprocessing configuration (error bars: ±1 std over 5 seeds).

- **Vocabulary.** Lowercasing and removing markup/numbers cut the vocabulary from 8,969 to 6,973 types (-22%), mostly by merging 'OIL'/'Oil'/'oil' and dropping figures. Stop-word removal barely changes the number of types but removes 42% of all tokens. Stemming shrinks the vocabulary further to 4,919 (lemmatization: 5,276, gentler); adding bigrams multiplies it to 35,285.
- **Classification.** The largest gain comes from stop-word removal (0.809 ± 0.045 → 0.852 ± 0.041): in a multinomial model, frequent function words contribute large, uninformative likelihood terms. Stemming or lemmatization adds about one more point (0.863 / 0.861) by pooling evidence across word forms, within one standard deviation. Bigrams did not help (0.848): with ~250 training stories most bigrams occur once and only add noise.
- **Clustering** is the most sensitive task. Normalising case and markup lifted purity from 0.557 to 0.648 (raw all-caps headlines and HTML copies otherwise form their own groups). Lemmatization gave the best purity and RI (0.665 ± 0.073, 0.833 ± 0.030). Bigrams hurt badly (0.558): more dimensions, sparser vectors and smaller distances between documents make the centroids less distinct. Differences among C-E are inside the seed-to-seed variation, so they should be read as trends, not firm rankings.
- **Recommendation** precision@5 is stable (0.71-0.73) across all settings: TF-IDF already down-weights common words, so the top neighbours are driven by rare, topical terms in every configuration. Preprocessing mainly improves the *explanations* (shared terms such as 'refinery, bpd' instead of 'the, said').
- **Choice:** configuration E (lemmatization) gives the best balance: top or near-top on every task, readable terms in the inverted index and explanations, and a vocabulary half the size of the raw one.

9.2 How do duplicates affect the results, and how were they detected and handled?

Detection. Exact duplicates: MD5 of lowercased, markup-free, punctuation-free text. Near duplicates: TF-IDF cosine ≥ 0.90 or word 5-shingle Jaccard ≥ 0.60. Pairs with high cosine but Jaccard < 0.35 are labelled *template*. Union-find merges pairs into groups. Result: 26 pairs - 2 exact (two genuine Reuters duplicates), 13 near (all 12 injected copies, i.e. recall 12/12, plus one real re-issue: 'PARTNERS CALL GENCORP RESPONSE UNPRODUCTIVE' sent twice with re-worded sentences), and 11 template pairs (five weekly FHLBB discount-note rate tables and two weekly API inventory bulletins).

Effect when duplicates are kept	Evidence
Classification: train/test leakage	With a naive split, test stories whose twin was in training were classified 100% correctly (58 cases over 10 splits) versus 85.5% for the others - the model memorises rather than generalises, inflating accuracy.
Recommendation: wasted slots	For 100% of the 30 documents that have a duplicate, the Top-1 recommendation was that duplicate (cosine 0.87-1.00), which tells the analyst nothing new. Precision@5 even looks better with duplicates kept (0.728 vs 0.721) because a copy always has the same label - a misleading gain.
Clustering: distorted centroids	100% of duplicate pairs land in the same cluster. Copies add weight to one story's vocabulary and templated groups can capture a whole centroid (cluster C2 in Section 6).

Mode	Docs	NB accuracy	NB macro-F1	Purity	Rand Index	P@5
Keep all (no de-duplication)	372	0.858 ± 0.026	0.861 ± 0.028	0.652 ± 0.045	0.818 ± 0.016	0.728
Remove exact + near duplicates	357	0.861 ± 0.035	0.862 ± 0.035	0.665 ± 0.073	0.833 ± 0.030	0.721
Also remove template duplicates	352	0.866 ± 0.026	0.865 ± 0.026	0.661 ± 0.051	0.842 ± 0.018	0.715

Handling. (1) One representative per exact/near group is kept (15 documents removed). (2) The train/test split is group-aware, so a story and its copy can never sit on opposite sides. (3) Template duplicates are kept for classification and clustering because they are distinct events with different figures, but they are flagged and excluded from a query's recommendation list. Removing them as well gives the highest Rand Index (0.842) and lowest variance, so a deployment could also collapse them into one 'series' item. The averaged accuracy differences between modes are small on this corpus because duplicates are only ~4-8% of it; the leakage measurement above shows the effect clearly where it occurs.

9.3 Recommendation approach: content-based vs collaborative

The application uses **content-based** recommendation because the corpus contains text and labels but no record of which analyst read, saved or forwarded which story. Each story's profile is its TF-IDF vector and similarity is cosine.

	Content-based (used)	Collaborative filtering
Data required	Only the documents (text, optional metadata)	A user x item interaction matrix: reads, clicks, ratings, dwell time
Cold start	New stories can be recommended immediately - essential for news	New stories have no interactions, so cannot be recommended; new users have no history
What it captures	Topical / lexical similarity; explainable by shared terms	Behavioural relatedness: 'analysts who read X also read Y', even with no shared words
Weaknesses	Over-specialisation (more of the same), no synonyms, no notion of quality or popularity	Sparsity, popularity bias, needs many active users, harder to explain
Preferable when	Items are new and short-lived, text is rich, few users (this desk)	Large stable user base and catalogue with long histories (e.g. research-report portal)

A **hybrid** is the natural next step for a real desk: content-based scores for fresh stories, blended with collaborative signals from the team's reading logs once they exist, plus a recency boost.

9.4 How do these techniques together help domain experts decide?

- **Routing and triage (classification):** Naive Bayes tags incoming stories with about 86% accuracy, so most stories reach the right desk automatically; low-confidence predictions (shown in the app) can be sent for a human check.
- **Theme discovery (clustering):** K-Means surfaced cross-desk themes the taxonomy hides - 'oil + tanker security' and 'trade friction + currency' - which is exactly where an FX or energy analyst should look for spill-over risk.

- **Noise reduction (duplicates):** collapsing re-issues and corrected copies reduces reading load and prevents a single story from being counted as several signals.
- **Context (recommendation):** opening a story shows related coverage with the shared terms that justify the link (e.g. the Bank of Japan intervention story links to Paris Accord statements by Miyazawa and Baker), giving the analyst background in one click.
- **Corpus analytics:** vocabulary, term and length statistics show what the feed is talking about and help tune domain stop-words (mln, dlrs, pct) and thresholds.

9.5 Naive Bayes vs K-Means, key findings, limitations and improvements

Using the appropriate metrics, Naive Bayes reaches accuracy 0.861 ± 0.035 and macro-F1 0.862 ± 0.035; K-Means reaches Purity 0.665 ± 0.073 and Rand Index 0.833 ± 0.030. The supervised model reproduces the editorial taxonomy far better; the unsupervised model recovers it partially and reveals where the taxonomy's classes overlap in language.

Key experimental findings

- Stop-word removal is the most valuable step for Naive Bayes; case/markup normalisation and lemmatization matter most for clustering; bigrams hurt both on a corpus this small.
- The confusions of both models coincide (interest/money-fx, crude/ship), so they reflect genuine topic overlap, not a modelling bug.
- K-Means is unstable: purity ranged roughly from 0.50 to 0.78 across seeds while the SSE varied by less than 2%. A lower SSE did not guarantee a partition closer to the taxonomy.
- K = 6 gave the best average purity (0.701) and RI in the K sweep (Figure 9), even though SSE keeps falling with K - the elbow is weak in sparse text.
- Duplicates cause train/test leakage (100% vs 85% accuracy) and fill recommendation lists with copies; group-aware splitting and filtering remove both problems.

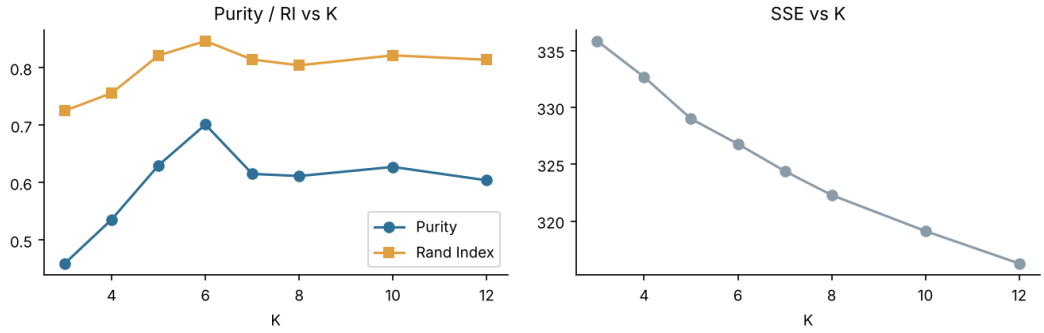


Figure 9 - Purity, Rand Index and SSE as a function of K (mean of 3 seeds).

Limitations

- Modest data: 357 stories (about 60 per topic) from 1987; vocabulary and topics differ from today's newswire.
- Single-label subset only: real Reuters stories often carry several topics (e.g. money-fx and interest together), which the single-label evaluation penalises.
- Bag-of-words features ignore word order, negation and meaning; TF-IDF cannot link synonyms.
- Recommendation is evaluated with a label-based proxy (precision@5), not with analyst relevance judgements.
- K-Means assumes roughly spherical, similar-sized clusters and is sensitive to seeding; Purity and RI favour some partitions artificially (e.g. Purity rises with K).

Suggested improvements for real-world deployment

- Train on the full Reuters/current feed with multi-label classification (one-vs-rest NB) and calibrated confidence thresholds for human review.

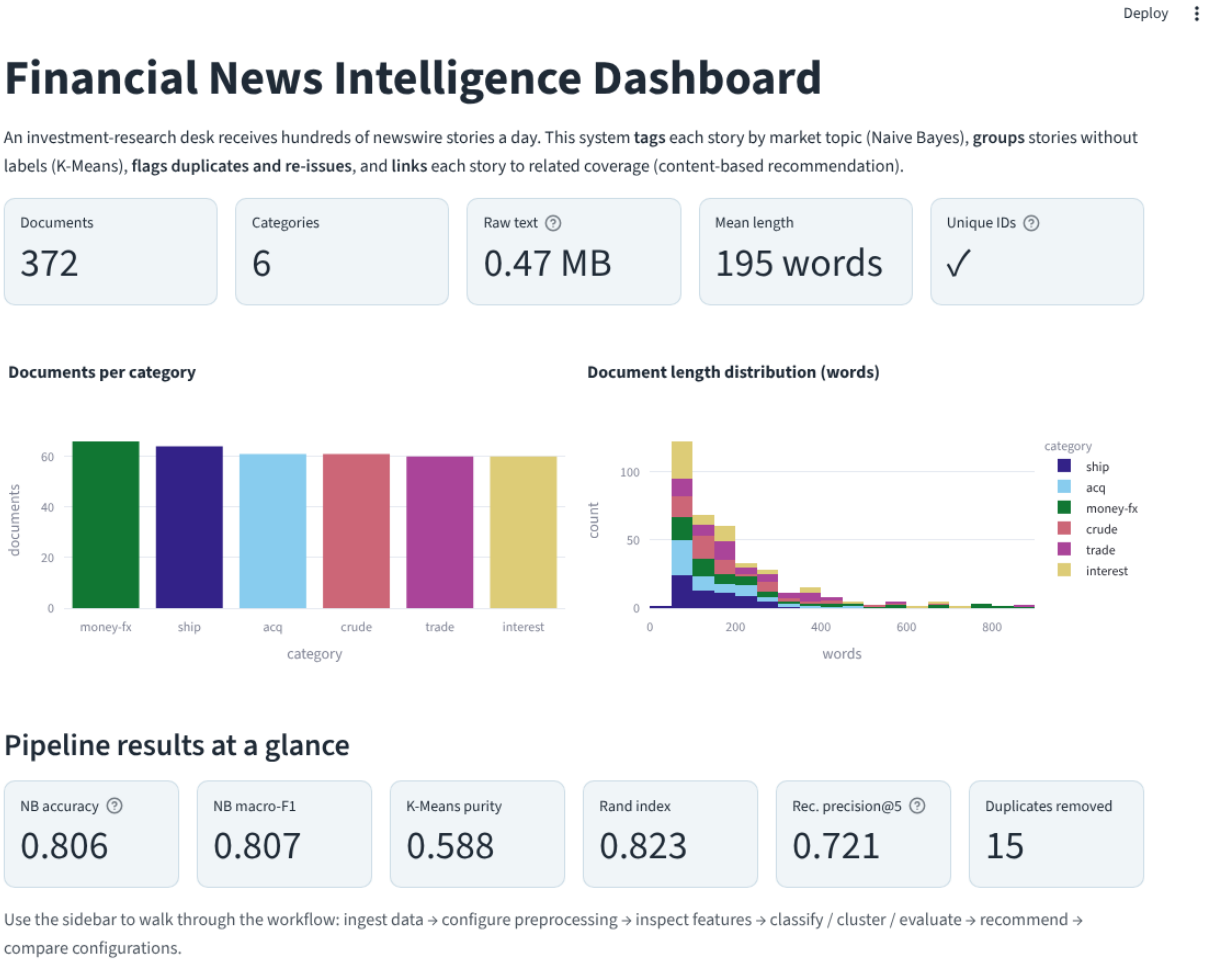
- Add dense sentence embeddings (or LSA) for clustering and recommendation to capture synonyms; keep TF-IDF for explanations.
- Use more K-Means restarts or spherical K-Means/bisecting K-Means, and report Adjusted Rand Index and NMI alongside Purity and RI.
- Run de-duplication incrementally (MinHash/LSH on shingles) at ingestion, and collapse templated series into one timeline item.
- Log analyst clicks and saves to enable a hybrid recommender, add recency and source weighting, and evaluate with real relevance feedback.
- Incremental index updates and scheduled retraining as the feed grows; monitor topic drift.

Appendix A - How to run

```
pip install -r requirements.txt
python -m nltk.downloader stopwords wordnet omw-1.4
streamlit run app.py
```

Optional: `python run_experiments.py` reproduces every number and figure in this report, and `python report/build_report.py` rebuilds this PDF.

Appendix B - Application screenshots



Screenshot - Dashboard and corpus overview

Deploy

Data ingestion

Source

(•) Bundled dataset (Reuters-21578 finance subset) ( ) Upload my own (CSV / JSON / ZIP of .txt)

Reload bundled dataset

Corpus statistics

Documents	Categories	Total words	Total size	Shortest / longest
372	6	72,681	0.47 MB	43 / 862 …

category	documents	mean_words	median_words	total_chars
acq	61	159	114	62604
crude	61	195.2	141	75469
interest	60	183.5	123.5	69505
money-fx	66	267.9	165	114517
ship	64	145.3	110.5	58980
trade	60	218.2	179.5	85660

Noise detected: 56 documents contain markup/HTML entities, 4 are mostly upper-case. Domain jargon (`mln` , `dlrs` , `pct` , ticker symbols like `<GY>`) is common.

Metadata (kept separate from text)

	doc_id	title	category	category_name	source	split	injected_c
0	RTR-07500	U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS	ship	Shipping	Reuters-21578 newswire (1987)	training	
1	RTR-11161	PARTNERS CALL GENCORP<GY> RESPONSE UNPRODUCTIVE	acq	Mergers & Acquisitions	Reuters-21578 newswire (1987)	training	
2	RTR-03622	CONTINENTAL HEALTH <CTHL> PURCHASES MARKETECH	acq	Mergers & Acquisitions	Reuters-21578 newswire (1987)	training	
3	RTR-03256	LONDON GRAIN FREIGHTS	ship	Shipping	Reuters-21578 newswire (1987)	training	
4	RTR-20311	BASTIAN TECHNOLOGIES MULLING COSMO <CSMO.O> BID	acq	Mergers & Acquisitions	Reuters-21578 newswire (1987)	test	
5	RTR-21343	FED SEEN ADDING RESERVES VIA SYSTEM REPOS	money-fx	Foreign Exchange	Reuters-21578 newswire (1987)	test	
6	RTR-05796	STUDY PREDICTS U.S. DEPENDENCE ON FOREIGN OIL	crude	Crude Oil	Reuters-21578 newswire (1987)	training	

Document text viewer

Document

RTR-07500 — U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS

U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS

`RTR-07500` · ship

Raw text

A U.S. Navy battle group led by the
U.S. aircraft carrier Kitty Hawk is in the northern Arabian Sea
amid renewed concern about the safety of shipping off the coast
of Iran, U.S. officials said today.
  But Pentagon spokesman Fred Hoffmann said that reports the
naval strike force was in the region did not mean the United
States was sending a new warning to Iran against escalating
attacks on shipping in the Persian Gulf.
  "Nothing like that has happened," he said. "No signal is being

Screenshot - Data ingestion and corpus statistics

Deploy

Text preprocessing & feature engineering

Preprocessing configuration

- [x] Lowercase
- [x] Strip HTML / markup
- [x] Remove numbers
- [x] Remove English stop-words
- [x] Remove newswire stop-words (said, mln, dlrs, pc…

Normalisation

- ( ) None
- ( ) Porter stemming
- (•) WordNet lemmatization

N-grams

- (•) Unigrams
- ( ) Uni + bigrams
- ( ) Up to trigrams

Minimum token length: 2

Min document frequency (index pruning): 2

Max document frequency (ratio): 0.90

Apply & rebuild features

Pipeline on one document

Example document

RTR-07500 — U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS

1. Raw text

```
A U.S. Navy battle group led by the
U.S. aircraft carrier Kitty Hawk is
amid renewed concern about the safet
of Iran, U.S. officials said today.
    But Pentagon spokesman Fred Hoff
naval strike force was in the region
States was sending a new warning to
attacks on shipping in the Persian G
    "Nothing like that has happened,
sent."
    He said that the Kitty Hawk is o
it usually does
```

2. Tokenised (before preprocessing)

```
A | U | S | Navy | battle | group | le
```

3. After preprocessing

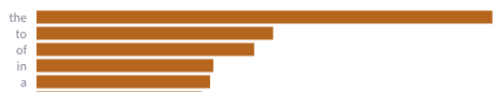
```
navy | battle | group | lead | aircraf
```

Vocabulary before vs after preprocessing

	Before (raw tokens)	After preprocessing	Change
Vocabulary size	8,793	5,276	-40.0%
Total tokens	71,061	37,798	-46.8%
Hapax legomena (freq = 1)	4,291	2,203	-48.7%
Mean tokens / doc	199.1	105.9	-46.8%
Type-token ratio	0.124	0.140	+12.8%

Inverted index keeps terms with df ≥ 2 and df ≤ 90% of documents: **2,609 indexed terms** (2,667 rare/ubiquitous terms pruned).

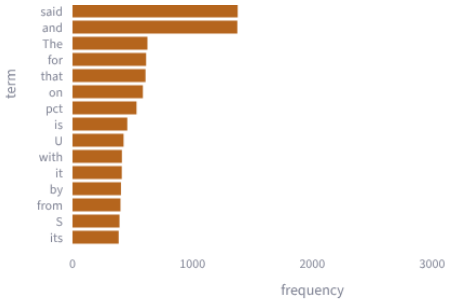
Top 20 terms — before

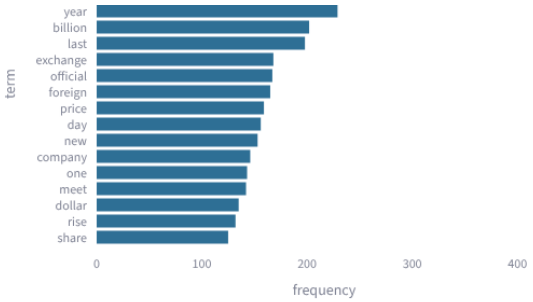


Top 20 terms — after

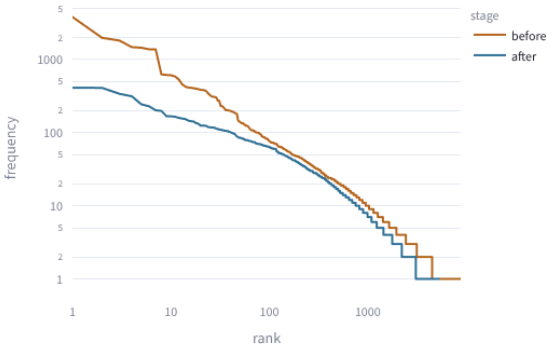


Screenshot - Preprocessing configuration and feature engineering (part 1/3)

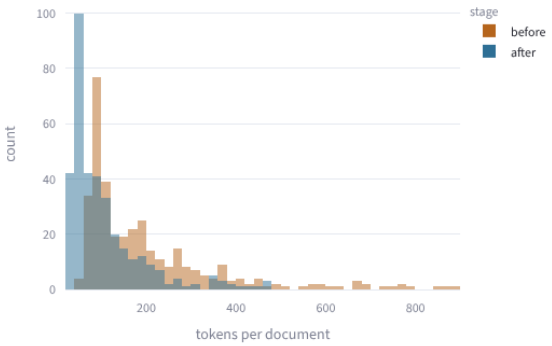




Rank-frequency (Zipf) distribution



Document length before / after



Inverted index explorer

	term	df	idf = log10(N/df)
0	market	130	0.4387
1	year	118	0.4808
2	last	118	0.4808
3	trade	114	0.4958
4	bank	113	0.4996
5	rate	112	0.5035
6	new	100	0.5527
7	tell	97	0.5659

Look up a term (after preprocessing)

oil

`oil` → df = **77**, idf = **0.6662**, postings (first 25):

	doc_id	title	tf	w = (1+log tf)·idf	normali
0	RTR-07500	U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS	1	0.6662	0
1	RTR-05796	STUDY PREDICTS U.S. DEPENDENCE ON FOREIGN OIL	9	1.3019	0
2	RTR-21131	AMPLE SUPPLIES LIMIT U.S. STRIKE'S OIL PRICE IMPACT	6	1.1846	0
3	RTR-02542	TALKS FAIL TO END BRAZILIAN SEAMEN'S STRIKE	1	0.6662	0
4	RTR-17875	SOUTH KOREA INCREASES DUTY ON CRUDE IMPORTS	2	0.8667	0
5	RTR-04125	CHINA CLOSES SECOND ROUND OF OFFSHORE OIL BIDS	4	1.0673	0
6	RTR-19285	U.S. OIL PRICES STRONG AHEAD OF OPEC MEETING	11	1.3599	0

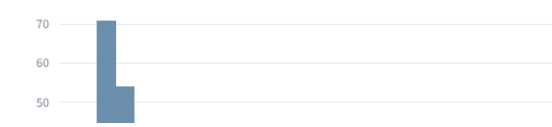
TF-IDF document vectors

$$w_{t,d} = \frac{(1 + \log_{10} tf_{t,d}) \cdot \log_{10} \frac{N}{df_t}}{\sqrt{\sum_{t'} \left[(1 + \log_{10} tf_{t',d}) \cdot \log_{10} \frac{N}{df_{t'}}\right]^2}}$$

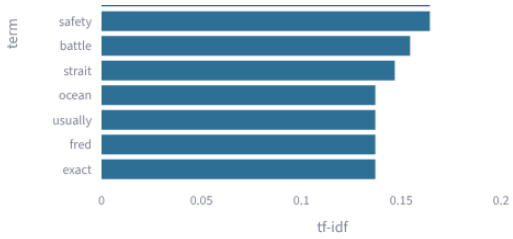
Top weighted terms in RTR-07500

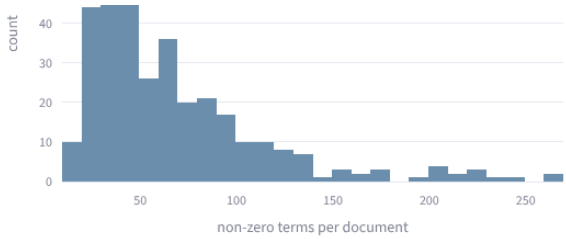


Feature density (sparsity 97.50%)



Screenshot - Preprocessing configuration and feature engineering (part 2/3)





Vector length before normalisation: 16.428; after: 1.000. Non-zero terms: 82.

Cosine similarity

Document A

RTR-07500

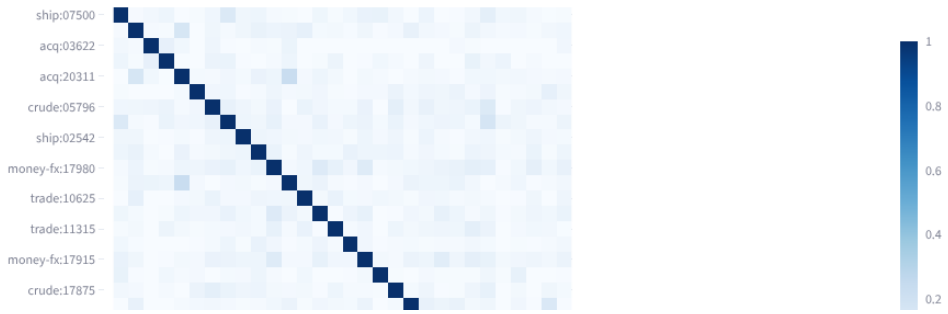
Document B

RTR-11161

cos(A, B)

0.0119

Cosine similarity heat-map (5 documents per category)



Screenshot - Preprocessing configuration and feature engineering (part 3/3)

Deploy

Text mining & corpus analytics hub

Model settings

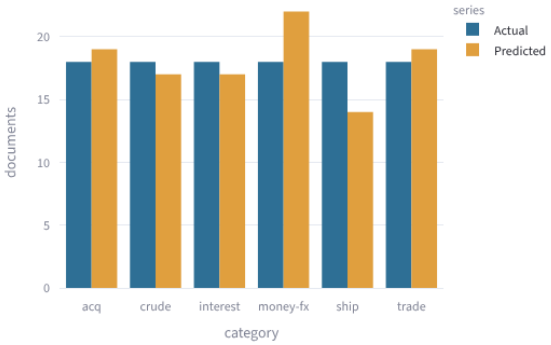
Naive Bayes classification | K-Means clustering | Evaluation & comparison | Duplicates

Multinomial Naive Bayes trained on **249** documents, tested on **108** (stratified split; duplicate groups kept on one side). Laplace smoothing α = 1.0.

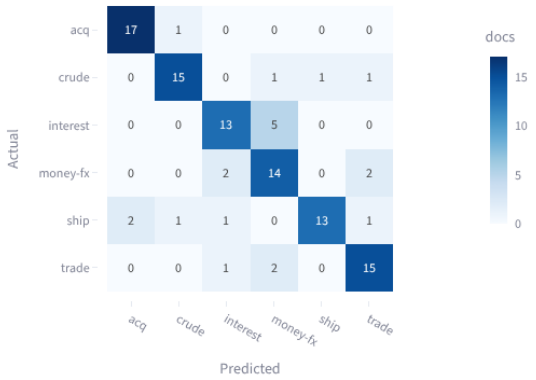
$$\hat{c} = \arg\max_{c} \left[\log P(c) + \sum_{t \in d} tf_{t,d} \log \frac{count(t,c) + \alpha}{\sum_{t'} count(t',c) + \alpha|V|} \right]$$

Accuracy	Macro precision	Macro recall	Macro F1	Weighted F1
0.806	0.816	0.806	0.807	0.807

Actual vs predicted class distribution (test set)



Confusion matrix



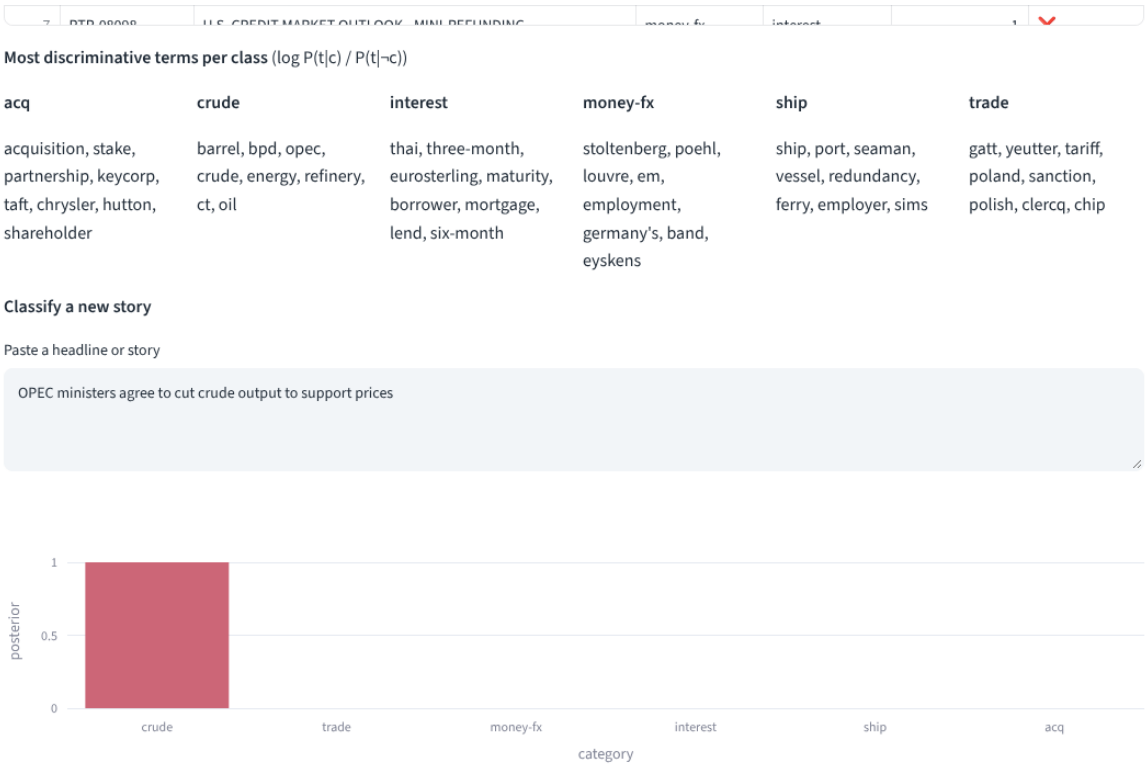
	precision	recall	f1	support
acq	0.895	0.944	0.919	18
crude	0.882	0.833	0.857	18
interest	0.765	0.722	0.743	18
money-fx	0.636	0.778	0.700	18
ship	0.929	0.722	0.813	18
trade	0.789	0.833	0.811	18
macro avg	0.816	0.806	0.807	108
weighted avg	0.816	0.806	0.807	108

Test-set predictions

☐ Show misclassifications only

	doc_id	title	actual	predicted	confidence	correct
0	RTR-07500	U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS	ship	ship	1	✅
1	RTR-05796	STUDY PREDICTS U.S. DEPENDENCE ON FOREIGN OIL	crude	crude	1	✅
2	RTR-01499	BALDRIGE CONCERNED ABOUT KOREAN/TAIWAN DEFICITS	trade	trade	1	✅
3	RTR-11738	LILLY <LLY> CUTS LIPOSOME CO <LIPO> STAKE	acq	acq	1	✅
4	RTR-10625	ISLAMIC BANKS ESTABLISH 50 MLN DLR TRADE PORTFOLIO	trade	money-fx	0.842	❌
5	RTR-04125	CHINA CLOSES SECOND ROUND OF OFFSHORE OIL BIDS	crude	crude	1	✅
6	RTR-06483	BANK OF ENGLAND LENDS ONE BILLION STG AT 10 PCT	interest	interest	1	✅

Screenshot - Text mining hub - Naive Bayes classification (part 1/2)



Screenshot - Text mining hub - Naive Bayes classification (part 2/2)

Deploy

Text mining & corpus analytics hub

› Model settings

Naive Bayes classification | K-Means clustering | Evaluation & comparison | Duplicates

K-Means with **K = 6** on L2-normalised TF-IDF vectors (squared Euclidean distance, which for unit vectors equals 2 − 2·cosine). Best of **5** restarts by SSE.

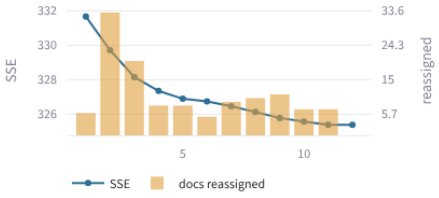
Step 1 — centroid initialisation (k-means++: each new seed sampled with probability ∝ squared distance to nearest seed)

	centroid	seed_doc	title	true_label
0	0	RTR-04425	MONTEDISON, OTHER REFINERIES TO SHUT FOR WORK	crude
1	1	RTR-20262	U.K. CREDIT POLICY SEEN STEADY, DESPITE BANK DATA	interest
2	2	RTR-12040	FHLBB CHANGES SHORT-TERM DISCOUNT NOTE RATES	interest
3	3	RTR-11349	KEYCORP <KEY> AGREES TO ACQUIRE UTAH BANK	acq
4	4	RTR-15384	GERMAN BANKERS' REMARKS REVIVE TALK OF RATE CUT	interest
5	5	RTR-20653	OUTER CONTINENTAL SHELF SALE POSTPONED	crude

Steps 2–3 — assign documents to nearest centroid, recompute centroids as cluster means

	iteration	reassigned	max_centroid_shift	sse	sizes
0	1	6	0.94172	331.645	[65, 88, 20, 58, 91, 35]
1	2	33	0.15543	329.697	[75, 87, 15, 53, 90, 37]
2	3	20	0.30468	328.138	[78, 90, 9, 52, 90, 38]
3	4	8	0.22278	327.337	[79, 91, 7, 50, 91, 39]
4	5	8	0.17046	326.880	[82, 87, 6, 49, 94, 39]
5	6	5	0.02609	326.735	[83, 88, 6, 49, 92, 39]
6	7	9	0.03902	326.454	[83, 83, 6, 49, 97, 39]

Convergence

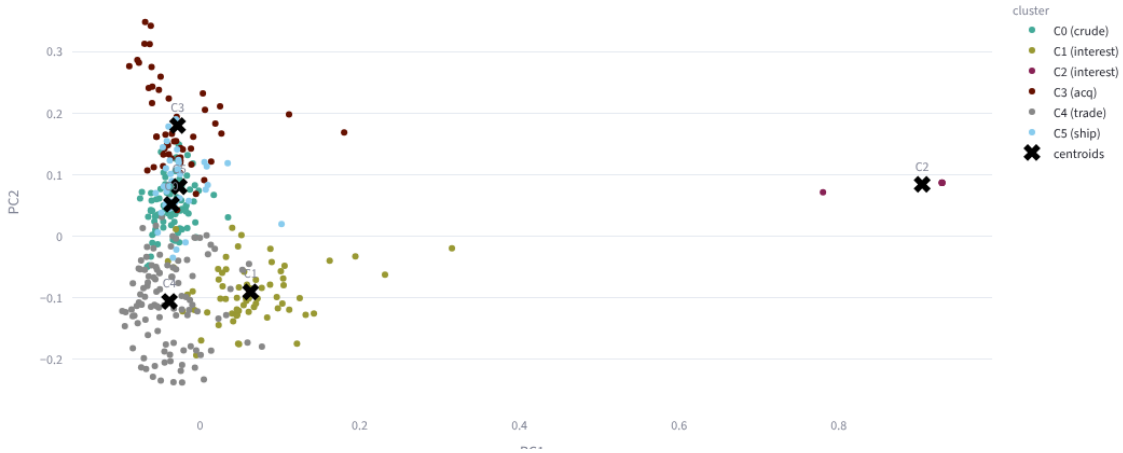


Step 4 — stopping criterion: stopped after 12 iterations because no document changed cluster (assignments converged). Restart SSEs: 330.56, 327.55, 325.37, 326.36, 329.67.

Colour points by

K-Means cluster | True category

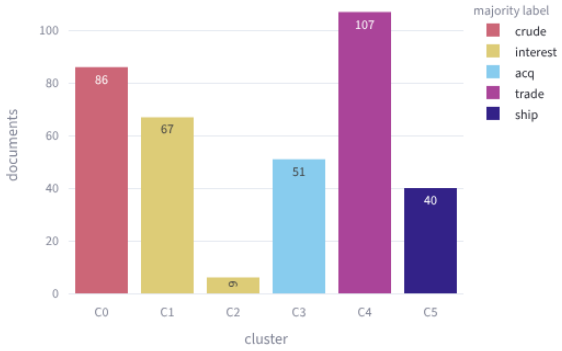
Documents in 2-D (PCA for display only; 3.2% of variance)



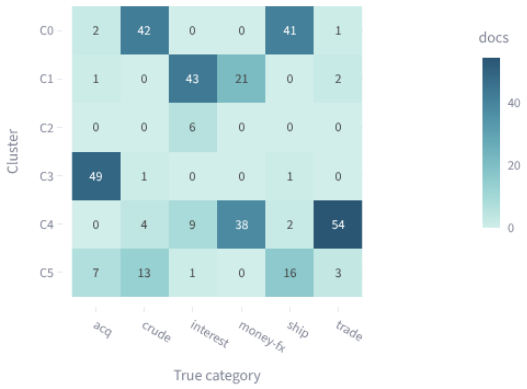
Screenshot - Text mining hub - K-Means clustering steps and visualisations (part 1/2)

PC1

Cluster sizes



Cluster × category contingency



Top terms per centroid

	cluster	size	majority	top_terms
0	C0	86	crude	oil, crude, barrel, ship, port, bpd, strike, opec, per, spokesman
1	C1	67	interest	bank, rate, stg, lend, money, prime, point, bill, market, compare
2	C2	6	interest	maturity, old, day, short-term, adjust, discount, loan, home, note, board
3	C3	51	acq	share, stake, stock, common, company, acquire, group, acquisition, outstanding, file
4	C4	107	trade	trade, japan, currency, economic, country, foreign, exchange, nation, tell, surplus
5	C5	40	ship	sea, water, canadian, continental, sink, drill, vessel, oil, ship, passenger

Cluster membership

Filter clusters

Choose options

	doc_id	title	true_label	cluster	cluster_majority
0	RTR-07500	U.S. SAYS NO SIGNAL BEING SENT BY SHIP MOVEMENTS	ship	C5	ship
1	RTR-11161	PARTNERS CALL GENCORP<GY> RESPONSE UNPRODUCTIVE	acq	C3	acq
2	RTR-03622	CONTINENTAL HEALTH <CTHL> PURCHASES MARKETECH	acq	C5	ship
3	RTR-03256	LONDON GRAIN FREIGHTS	ship	C5	ship
4	RTR-20311	BASTIAN TECHNOLOGIES MULLING COSMO <CSMO.O> BID	acq	C3	acq
5	RTR-21343	FED SEEN ADDING RESERVES VIA SYSTEM REPOS	money-fx	C1	interest
6	RTR-05796	STUDY PREDICTS U.S. DEPENDENCE ON FOREIGN OIL	crude	C5	ship
7	RTR-21131	AMPLE SUPPLIES LIMIT U.S. STRIKE'S OIL PRICE IMPACT	crude	C0	crude

Screenshot - Text mining hub - K-Means clustering steps and visualisations (part 2/2)

Deploy

Text mining & corpus analytics hub

› Model settings

Naive Bayes classification | K-Means clustering | Evaluation & comparison | Duplicates

Purity and Rand Index are computed with hand-written functions over cluster assignments and true labels.

$$\text{Purity} = \frac{1}{N}\sum_{k}\max_{j}|\omega_k \cap c_j| \qquad RI = \frac{TP+TN}{TP+FP+FN+TN}$$

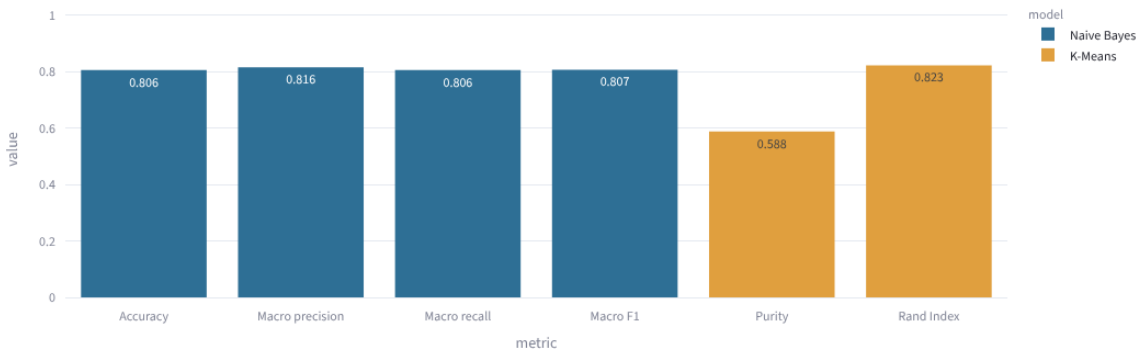
	same class	different class
same cluster	6386	4057
different cluster	7221	45882

63,546 document pairs = N(N−1)/2 with N = 357.

	cluster	size	majority count	majority label
0	C0	86	42	crude
1	C1	67	43	interest
2	C2	6	6	interest
3	C3	51	49	acq
4	C4	107	54	trade
5	C5	40	16	ship

Supervised (Naive Bayes) vs unsupervised (K-Means)

Evaluation metrics



Interpretation. Naive Bayes reaches **80.6% accuracy** (macro-F1 0.807) because it learns the label definitions from 249 labelled stories; its weakest class is **money-fx** (F1 0.70). K-Means sees no labels and reaches **purity 0.588** — well above the 16.8% you would get by putting everything in one cluster — and **Rand Index 0.823**. The Rand Index is inflated by the many 'different class, different cluster' pairs (TN = 45,882); a random 6-way split would already score about 0.72. Impure clusters: C0 mixes crude + ship; C1 mixes interest + money-fx; C4 mixes trade + money-fx; C5 mixes ship + crude — these topics share vocabulary (e.g. rates and currencies, or tankers and oil), so an unsupervised grouping by word usage does not match the editorial taxonomy exactly. The two are complementary: NB is the right tool when a stable taxonomy exists; K-Means helps discover themes and sanity-check the taxonomy.

Screenshot - Text mining hub - evaluation and comparison

Deploy

Text mining & corpus analytics hub

› Model settings

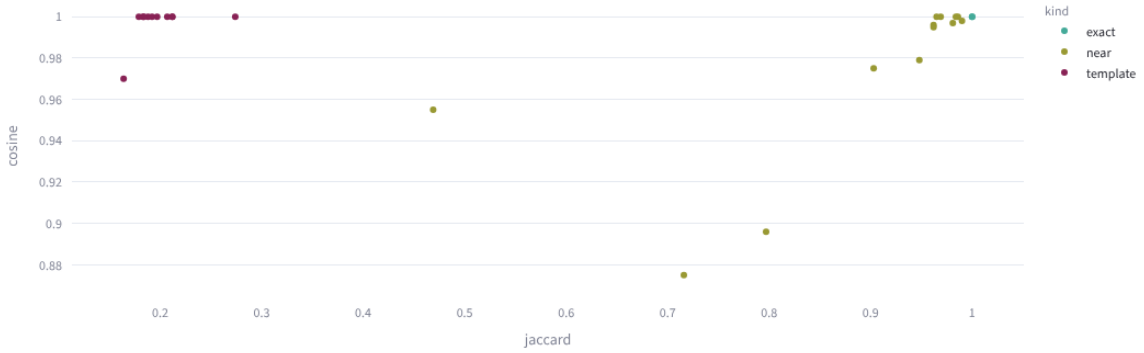
Naive Bayes classification | K-Means clustering | Evaluation & comparison | Duplicates

Duplicate pairs	Exact	Near (re-issued)	Template	Removed from analysis
26	2	13	11	15

Detection: MD5 hash of normalised text (exact), word 5-shingle **Jaccard** and TF-IDF **cosine** (near). Duplicates are grouped with union-find; one representative per group is kept, and during the train/test split a whole group always goes to the same side to prevent leakage.

	doc_a	title_a	doc_b	title_b	cosi
8	RTR-08710	SUMITA SAYS BANK WILL INTERVENE IF NECESSARY	RTR-08658	SUMITA SAYS BANK WILL INTERVENE IF NECESSARY	
14	RTR-11817	H.K. BANKS TO RAISE PRIME RATES SOON, DEALERS SAY	RTR-11848	H.K. BANKS TO RAISE PRIME RATES SOON, DEALERS SAY	
1	RTR-07950	G-10 FINANCE OFFICIALS DISCUSS DEBT, CURRENCIES	DUP-008	G-10 Finance Officials Discuss Debt, Currencies	
7	RTR-18160	NEED FOR BIGGER ALLIED FORCE IN GULF PLAYED DOWN	DUP-011	Need For Bigger Allied Force In Gulf Played Down	
18	DUP-005	Bank Of Finland Will Announce New Measures	RTR-07568	BANK OF FINLAND WILL ANNOUNCE NEW MEASURES	
22	DUP-002	High Winds Keep Vessels Trapped In Baltic Ice	RTR-07642	HIGH WINDS KEEP VESSELS TRAPPED IN BALTIC ICE	
23	RTR-11587	U.S. CARGO PREFERENCE SQUABBLE CONTINUES	DUP-012	CORRECTED - U.S. CARGO PREFERENCE SQUABBLE CONTINUES	0.9
2	DUP-009	CORRECTED - TREASURY'S BAKER SUPPORTS FED MONETARY POLI	RTR-07669	TREASURY'S BAKER SUPPORTS FED MONETARY POLICY	0.9
4	RTR-08234	NIGERIA CHANGES AUCTION RULES TO DEFEND NAIRA	DUP-006	CORRECTED - NIGERIA CHANGES AUCTION RULES TO DEFEND NAI	0.9
24	DUP-003	CORRECTED - N.Z. PORTS REOPEN, BUT FURTHER DISRUPTION L	RTR-08214	N.Z. PORTS REOPEN, BUT FURTHER DISRUPTION LIKELY	0.9

Duplicate pairs: shingle overlap vs TF-IDF cosine



Recall on the 12 injected near-duplicates: 12/12.

Screenshot - Text mining hub - duplicate detection

Deploy

Document recommendation

Method: content-based Top-K. Every story is a length-normalised TF-IDF vector; related stories are those with the highest cosine similarity. Free-text queries are scored term-at-a-time through the inverted index.

Query by

Select a document Enter free text

Top-K

5

Hide duplicates of query

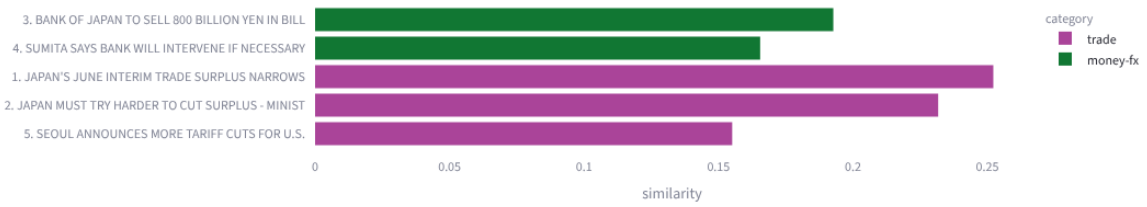
Describe what you are looking for

Japan trade surplus with the United States and yen intervention

Inverted-index scoring touched 176 candidate documents out of 357.

rank	doc_id	title	category	similarity	source	shared_terms	same_category
1	RTR-19044	JAPAN'S JUNE INTERIM TRADE SURPLUS NARROWS	trade	0.2522	Reuters-21578 newswire (1987)		—
2	RTR-19062	JAPAN MUST TRY HARDER TO CUT SURPLUS - MINISTER	trade	0.2317	Reuters-21578 newswire (1987)		—
3	RTR-16177	BANK OF JAPAN TO SELL 800 BILLION YEN IN BILLS	money-fx	0.1927	Reuters-21578 newswire (1987)		—
4	RTR-08710	SUMITA SAYS BANK WILL INTERVENE IF NECESSARY	money-fx	0.1655	Reuters-21578 newswire (1987)		—
5	RTR-17511	SEOUL ANNOUNCES MORE TARIFF CUTS FOR U.S.	trade	0.1551	Reuters-21578 newswire (1987)		—

Recommendation scores



How it works & limitations

Screenshot - Recommendation from a free-text query

Deploy

Performance analytics

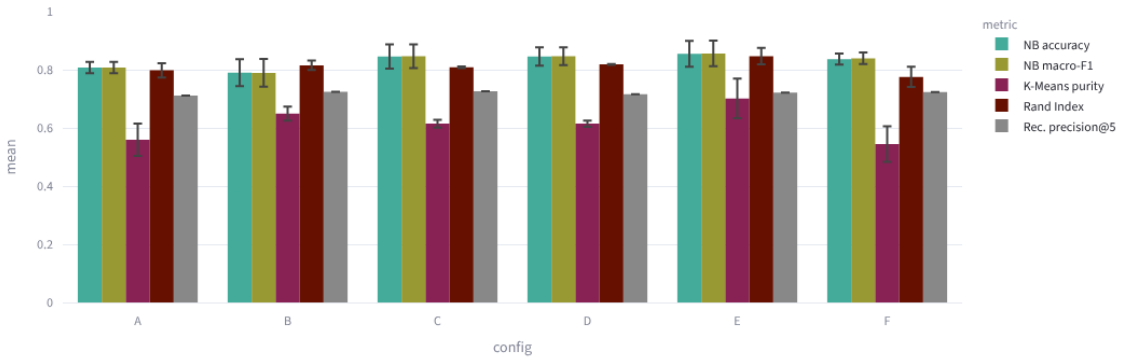
Controlled experiments that re-run the complete pipeline under different settings. Each figure is the mean over 3 random seeds (train/test split + K-Means seeding); error bars show ±1 std.

Preprocessing comparison | Duplicates | Choice of K

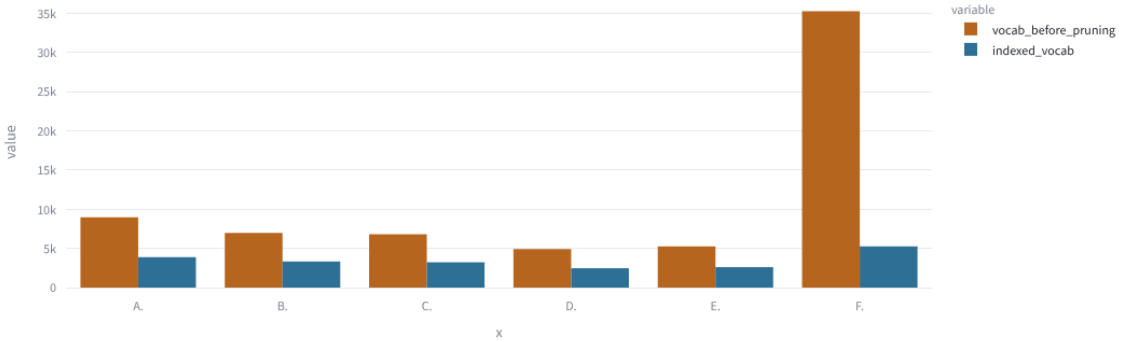
Run preprocessing experiment (≈1 min)

	config	vocab_before_pruning	indexed_vocab	tokens	nb_accuracy	nb_macro_f1	km_purity	km_rand_index	rec_pr
0	A. Raw tokens (no cleaning)	8969	3885	71642	0.807	0.807	0.560	0.798	
1	B. Lowercase + markup/numbers removed	6973	3349	65521	0.790	0.789	0.649	0.815	
2	C. B + stop-words (generic + domain)	6822	3217	37811	0.846	0.847	0.614	0.808	
3	D. C + Porter stemming	4919	2477	37811	0.846	0.847	0.614	0.818	
4	E. C + WordNet lemmatization	5276	2609	37798	0.855	0.856	0.701	0.847	
5	F. E + bigrams	35285	5270	75239	0.836	0.839	0.544	0.775	

Effect of preprocessing (A = raw … F = lemma + bigrams)



Vocabulary size per configuration



A. Raw tokens (no cleaning) — unigrams

B. Lowercase + markup/numbers removed — lowercase, strip markup, drop numbers, unigrams

C. B + stop-words (generic + domain) — lowercase, strip markup, drop numbers, stop-words, domain stop-words, unigrams

D. C + Porter stemming — lowercase, strip markup, drop numbers, stop-words, domain stop-words, Porter stemming, unigrams

Screenshot - Performance analytics - preprocessing comparison